



September 10, 2026 at 10:41 PM

CSCI 104 HW 1

Problem 1

```

a) void f, (int n)
    {
        int i = 2   $\Theta(1)$ 
        while (i < n) {
            i = i * i;
        }
    }

```

$\checkmark$
 $i = i^2$

ite. $\rightarrow i = i^2$

0 $\rightarrow$ 2 (start)

1 $\rightarrow 2^2 = 4$

2 $\rightarrow 4^2 = 16$

3 $\rightarrow 16^2 = 256$

4 $\rightarrow 256^2$

$\rightarrow 2^k$

}
}

$$i = i^2$$

4

256

$$2^{2^k}$$

$$T(n) = \Theta(1) + O\left(\sum_{j=0}^{n-1} \Theta(1)\right) \checkmark$$

$$= \Theta(1) + O(\log_2(\log_2 K))$$

$$= \Theta(1) + \Theta(\log \log n)$$

$$\boxed{= \Theta(\log \log n)}$$

```
b) void f2(int n)
{
    for (int i=1; i<=n; i++) {
        if ( (i % (int) sqrt(n)) == 0 ) {
            for (int k=0; k<pow(i,3); k++) {
            }
        }
    }
}
```

$$T(n) = \sum_{i=1}^n \sqrt{n} \quad \overset{\text{pow}(i,3)}{\sum_{k=0} \Theta(1)}$$

$$= \Theta(\sqrt{n}) * \Theta(n^3)$$

$$= \boxed{\Theta(n^{3/2})}$$

```

C. for (int i=1; i<=n; i++) {
    for (int k=1; k<=n; k++) {
        if (A[k] == i) {
            for (int m=1; m<=n; m=m+m) {
            }
        }
    }
}
    
```

$$T(n) = \sum_{i=1}^n \left(\sum_{k=1}^n (\Theta(1)) \right) + \sum_{m=1}^{\log_2 n} \Theta(1)$$

$$= \Theta(n) * \Theta(n) + \Theta(\log n)$$

$$= \Theta(n^2) + \Theta(\log n)$$

$$= \boxed{\Theta(n^2)}$$

d) int f(int n)

{

int *a = new int[10];

int size = 10

```
int *a = new int [10];
```

```
int size = 10
```

```
for (int i = 0; i < n; i++)
```

```
{
```

```
    if (i == size)
```

```
    { int newSize = 3 * size / 2
```

```
      int *b = new int [newSize];
```

```
      for (int j = 0; j < size; j++)
```

```
        b[j] = a[j];
```

```
      delete [] a;
```

```
      a = b;
```

```
      size = newSize;
```

```
    }
```




Ad



Search



Size - no size

$$a[i] = t \neq i;$$

$$\{ \}$$

$$\Theta(1) + \Theta(1) + \sum_{i=1}^n \Theta(1) * \Theta(1) * \Theta(1) * \sum_{j=0}^{\text{size}} \Theta(1)$$

$$T(n) = \Theta(n) * \Theta(1)$$

$$= \boxed{\Theta(n)}$$



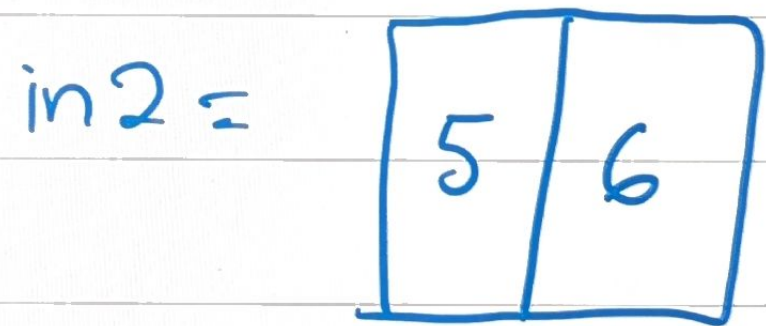
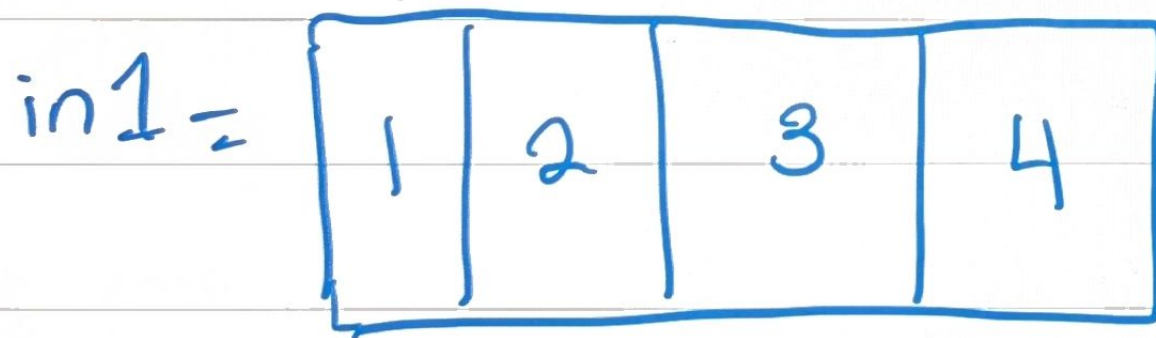
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Problem 2

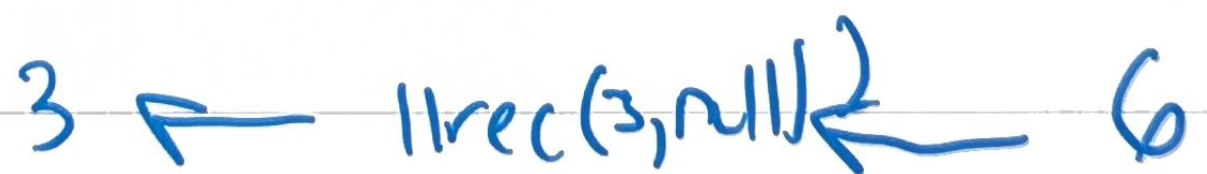
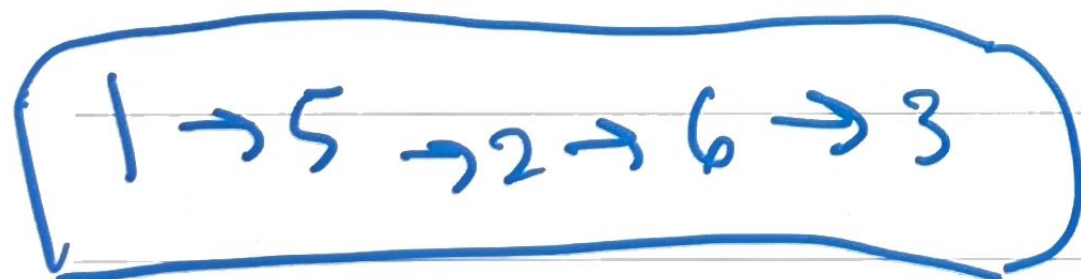
a) $in1 = 1, 2, 3, 4$ $in = 5, 6$ 
 $llrec(1, 5) \rightarrow llrec(5, 2) \rightarrow llrec(2, 6) \rightarrow$

R:

1

5

2

 $llrec(6, 3)$ b) $in1 = null$ $in2 = 2$



Ad

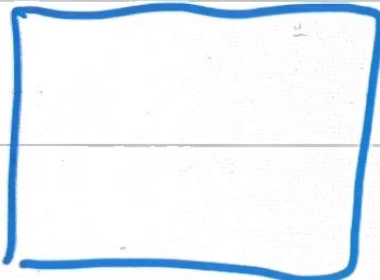


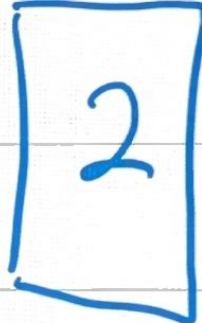
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b) $in1 = \text{nullptr}$

$in2 = 2$

$in1 =$ 

$in2 =$ 

↳ if ($in1 = \text{nullptr}$)
return $in2$

1st B.C



2 is returned